

Revision Questions

1. Define each of the following

a) Byte

8 Bits

b) Nibble

4 bits

c) Binary image

pixel = 1 bit

1 bit-monochrome

contains no color

d) Fixed Point Numbers

fixed(w,b)

w=>width of number

b =>binary point position

2. Compare Between

Binary Image – Gray Image – Color Image

binary image

pixel = 1 bit

1 bit-monochrome

contains no color

gray image

pixel = 8 bit

0 to 255

256 value

color image

pixel = 24 bit

red, green, blue

$256 * 256 * 256$

3. Convert the decimal number $(255)_{10}$ to its binary equivalent.

.....	
255/2=127.5	1
127/2=63.5	1
63/2=31.5	1
31/2=15.5	1
15/2=7.5	1
7/2=3.5	1
3/2=1.5	1
1/2=0.5	1
.....	
.....	
11111111	
.....	
.....	
.....	

4. Convert the octal number $(275)_8$ to its decimal equivalent.

$$5 \cdot 1 + 7 \cdot 8 + 2 \cdot 8 \cdot 8 = 189$$

5. Convert the hexadecimal number $(ADF5)_{16}$ to its binary equivalent.

$$1010 \ 1101 \ 1111 \ 0101$$

6. Convert the octal number $(45)_8$ to its binary equivalent.

$$100 \ 101$$

7. Convert the hexadecimal number $(FABB)_{16}$ to its Octal equivalent.

.....
.....
1111 1010 1011 1011
.....

.....
1 111 101 010 111 011
.....

.....
1 7 5 2 7 3
.....
.....

8. Converting the following decimal floating point number

$(12.75)_{10}$ to its binary equivalent.

$$12/2=6$$

$$6/2=3$$

$$3/2=1.5$$

$$1/2=0.5$$

1100

$$0.75*2=1.5$$

$$0.5*2=1$$

1100.11

9. Converting the following hexadecimal number $(FACB)_{16}$ to its decimal equivalent.

$$\begin{aligned} & 11 \cdot 1 \\ & + \\ & 12 \cdot 16 \\ & + \\ & 10 \cdot 16 \cdot 16 \\ & + \\ & 15 \cdot 16 \cdot 16 \cdot 16 \\ & = 64203 \end{aligned}$$

10. Converting the following binary floating $(1011.101)_2$ to its decimal equivalent.

.....

.....

$$1+2+0+8=11$$

.....

.....

$$0.5+0.125=0.625$$

.....

.....

$$=11.625$$

.....

12. Converting the following decimal floating point number
(0.18)₁₀ to its binary equivalent.

.....
0.18*2=0.36.....
0.36*2=0.72.....
0.72*2=1.44.....
0.44*2=0.88.....
0.88*2=1.76.....
0.76*2=1.52.....
0.52*2=1.04.....
0.04*2=0.08.....
0.08*2=0.16.....
.....
.....

Revision Questions

1. Demonstrate the differences between computer architecture and computer organization.

.....
computer architecture.....

.....
is the abstract model and is the
programmer's view in~ terms of
instructions, addressing modes and
registers.
.....

.....
describes what the computer does
.....

.....
indicates computer hardware.....
.....

.....
architecture is fixed first when designing
computers
.....

.....
computer organization.....

.....
expresses the realization of the architecture
.....

.....
describes how it does it
.....

.....
Reveals computer performance
.....

.....
Is decided after the architecture is fixed
.....

3. List the ALU registers clarifying each register usage.

Memory Buffer Register (MBR): Contains the word to be stored in memory or sent to the I/O unit, or is used to receive a word from memory or from the I/O unit.

Memory Address Register (MAR): Specifies the address in memory the word to be written from or read into the MBR.

Instruction Register (IR): Contains the 8-bit op-code instruction being executed.

Instruction Buffer Register (IBR): Employed to hold temporarily the right-hand instruction from a word in memory.

— Program Counter (PC): Contains the address of the next instruction-pair to be fetched from memory.

Accumulator (AC) : holds
ø output of the ALU after an arithmetic operation
* temporarily operands loaded from memory
ø the most-significant digits of a product
ø the divisor for division

Multiplier Quotient (MQ): holds
ø temporarily operands
ø results of ALU operations
ø least-significant bits of the product as multiplication proceeds¹⁷¹
* quotient from division

Data per Location: bits (2 bytes)

5. Demonstrate the differences between synchronous and asynchronous buses.

Synchronous Bus

Contains a clock (CLK) that sends out a sequence of 1's and 0's at timed intervals to synchronize bus operations.

ADV Lower overhead and greater throughput

DIS Slightly more complex
- Hardware is expensive

Asynchronous Bus

¢ No clock signal
¢ Transmitters and receivers in all timing based on control signal edges

ADV Simple; doesn't require synchronization of both communication sides

Cheap, because Asynchronous transmission require less hardware

Set-up is faster than other transmissions, so well suited for applications where messages are generated at irregular intervals,

DIS Large relative overhead,

6. Describe in details the control buses operations.

Carries the signals from the CU relating to the control and co-ordination of activities across computer.

8. List in details the interrupts types.

Interrupt Types

External Interrupts

— come from external input / output devices which are connected externally to the processor.

Internal Interrupts

— known as traps

— Caused by

- * some illegal operation
- * erroneous use of data.
- * exception that has been caused by the program itself.
 - E.G. : division by zero, processor timer or an invalid op-code

Software interrupts

— occur only during the execution of an instruction.

◊ E.G: switch from user mode to supervisor mode.

10. Consider a hypothetical microprocessor generating a 16-bit address (for example assume that the program counter and the address registers are 16 bits wide) and having a 16-bit data bus

- a) What is the maximum memory address space that the processor can access directly if it is connected to a "16-bit memory"?
- b) What is the maximum memory address space that the processor can access directly if it is connected to an "8-bit memory"?
- c) What architectural features will allow this microprocessor to access a separate "I/O space"?
- d) If an Input and an output instruction can specify an 8-bit I/O port number, how many 8-bit I/O ports can the microprocessor support? How many 16-bit I/O ports? Explain.

Addressable Locations: The processor still generates 2^{16} unique addresses.

$$2^{16} = 65,536 \text{ locations}$$

Data per Location: 16 bits (2 bytes).

Total Capacity: $65,536 \text{ locations} * 2 \text{ bytes} = 131,072 \text{ bytes}$

B

$2^{16} = 65,536$ locations

Data per Location: 8 bits (1 bytes)

Total Capacity = $65,536 * 1 = 65,536$ bytes

D 256 Input/Output Port

$2^8 = 256$ Ports

Data per Location: bit=s (2 bytes)

11. Define each of the following

a) Bus Arbitration

more than one module may need control of the bus,

.....
.....
.....
.....
.....

b) Daisy Chain

I/O modules share a common interrupt request lines.
a word (vector) on the data lines.
The vector is either the address of the I/O module or some other unique
identification.
priority of a module is determined by the position of the module in the daisy
chain
the processor uses the vector as a pointer

c) Software poll

.....
.....

Revision Questions

1. Define each of the following

a) access type

the action that physically takes place, during a read or write operation.

b) cycle time

time elapsed from the start of a read operation to the start of a subsequent read.

c) latency

time interval between the request for information and the access to the first bit of that information.

d) Cache memory

a small amount of fast memory placed between two levels of memory hierarchy to bridge the gap in access times between processor and main memory and between main memory and disk.

e) Cache miss

If the request corresponds to an element that is not currently in the cache,

Not found in cache

f) Cache hit

If the request corresponds to an element that is currently residing in the cache

found in cache

g) Miss penalty

the time to copy data from main memory to the cache.

h) Average memory access time

$$AMAT = \text{Hit Time} + (\text{Miss Ratio} \times \text{Miss Penalty})$$

2. Discuss the differences between SRAM and DRAM.

SRAM	DRAM
Expensive	cheap
Need no refresh	Need to be refreshed
Shorter access time	Longer access time
Smaller amount of data	Larger amount of data
	SDRAM,DDR SDRAM

3. List briefly the CPU registers types.

.....
.....
.....
General Purpose Registers: used to store data and intermediate results during program execution. Its contents can be accessed through assembly programming, as AX, BX, CX and DX registers.
.....
.....
.....
.....
.....

* Special Purpose Registers: Users do not access these registers. These are used by computer system at the time of program execution as MAR, MBR, IR and PC registers
.....
.....
.....

* Segment Registers: as CS, DS and SS registers.
.....
.....
.....

* Special Kernel Mode Registers: are intended for writing operating systems, debuggers, and other system level tools.
.....
.....
.....
.....
.....

4. Clarify the differences between Direct Mapped cache, Fully Associative cache and Set Associative cache.

.....

Mapping technique	Simplicity	Associative tag search	Expected cache utilization	Replacement technique
Direct	Yes	None	Low	Not needed
Associative	No	Involved	High	Yes
Set-associative	Moderate	Moderate	Moderate	Yes

.....

5. Discuss the differences between the three replacement policies
enhancing your answer with clarifying block diagrams.

Least Recently Used (LRU)

When performing LRU caching, you always throw out
the data that was least recently used.

Random Replacement

» select a block to replace in a totally random order, with
no regard to memory references or previous selections.

FIFO

It removes the block that has been in the cache for the
longest time.

6. Design a $64\text{M} \times 8\text{-bit}$ memory using a number of $16\text{M} \times 1\text{-bit}$ DRAM chips. Assume that each individual chip has a chip select ($\overline{CS}$) line and a read/write (R/W) line. Compute the number of chips required and show a complete connection diagram of the designed memory.

rows number of word = $64\text{M}/16\text{M}=4$

Cloumns data line = $8/1=8$

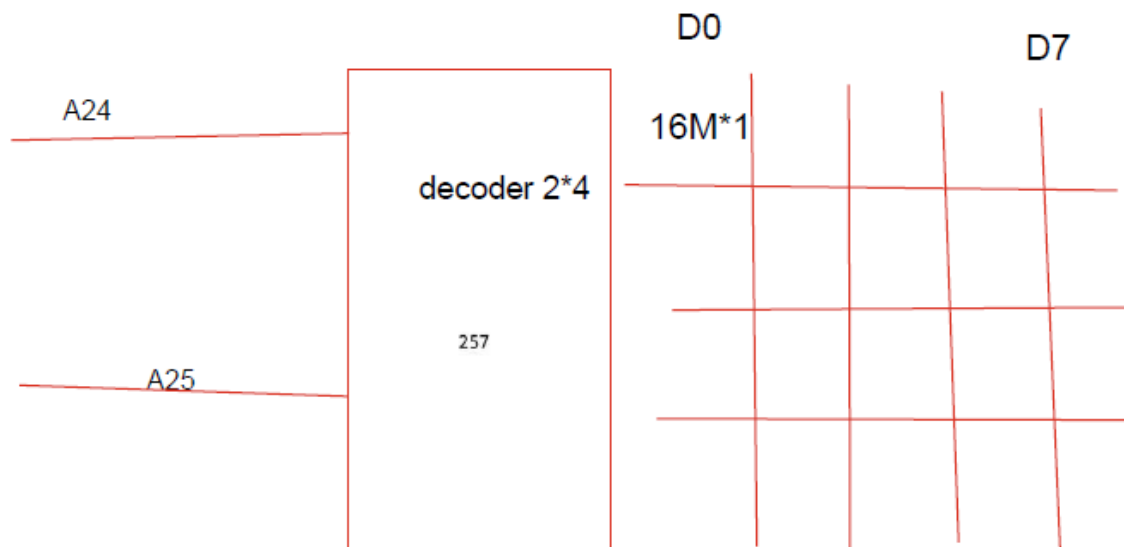
number of chips = $4 \times 8 = 32$ chips

target memory = $2^{26} = (A0 \Rightarrow A25)$

using memory = $2^{24} = (A0 \Rightarrow A23)$

difference = $26 - 24 = 2$ lines ($A24 \Rightarrow A25$)

decoder = 2×4 input 2 output $2^2 = 4$



7. The memory units that follow are specified by the number of words times the number of bits per word. How many address lines and input-output data lines are needed in each case?

a) $8K \times 16$

Data lines = 16 input-output
number of word = 8K = 2^{13}
number of address lines = 13

b) $2G \times 8$

Data lines = 8 input-output
number of word = 2×2^{30}
number of address lines = 31

c) $16M \times 32$

.....
.....
Data lines = 32 input-output
number of word = $2^4 \times 2^{20} = 2^{24}$
number of address lines = 24
.....
.....
.....

d) $256K \times 64$

.....
.....
Data lines = 64 input-output
number of word = $2^8 \times 2^{10} = 2^{18}$
number of address lines = 18
.....
.....

8. Using 64×8 ROM chips with an enable input, construct a 512×8 ROM with eight chips and a decoder.

$$512 \times 8 / 64 \times 8 = 8 \times 1$$

$$2^9 \times 1$$

$$2^6 \times 1$$

rows = 8
columns = 1
chip = $8 \times 1 = 8$

address line target = 9
address line small = 6
difference = $9 - 6 = 3$ ($A_6 \Rightarrow A_8$)
decoder = $3 \times 2^3 = 8$

9. Design a memory of total capacity 256 Kilo Byte using DRAM chips of size $256 \times 8\text{bit}$. Give the array configuration of the chips on the memory board showing all required input and output signals.

$$\text{target memory} = 2^8 \times 2^{10} = 2^{18} \\ 2^{18} \times 8\text{bit}$$

using memory $2^8 \times 8\text{ bit}$

$$\text{row} = 2^{18} / 2^8 = 2^{10} = 1024$$

$$\text{columns} = 8 / 8 = 1$$

$$\text{number of chips} = 1024 \times 1 = 1024$$

$$\text{address line target memory} = 2^{18} = (A0 \Rightarrow A17)$$

$$\text{address line small memory} = 2^8 = (A0 \Rightarrow 7)$$

$$\text{difference address line} = 18 - 8 = 10 \quad (A8 \Rightarrow A17)$$

$$\text{decoder} = 10 \times 1024$$

